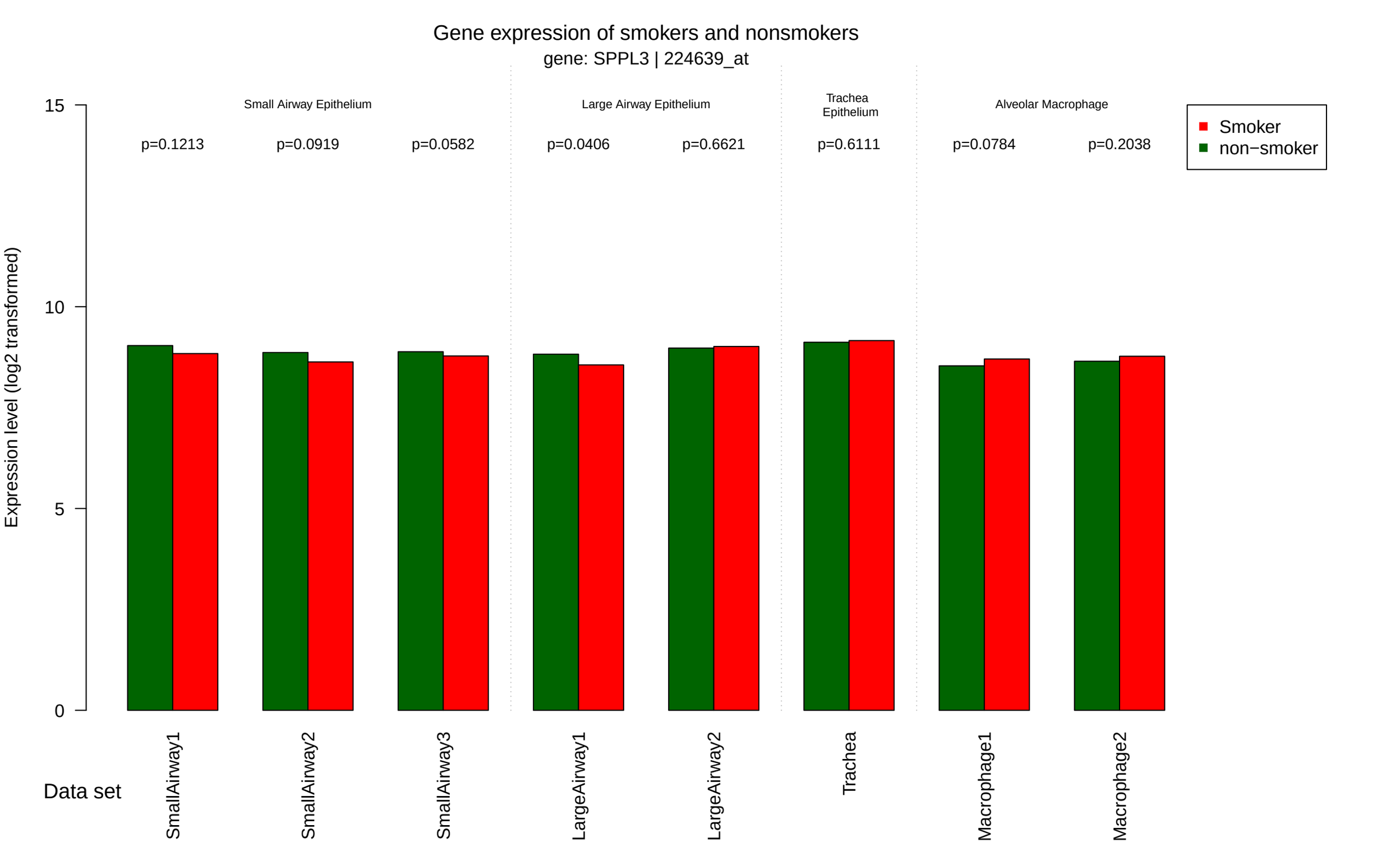
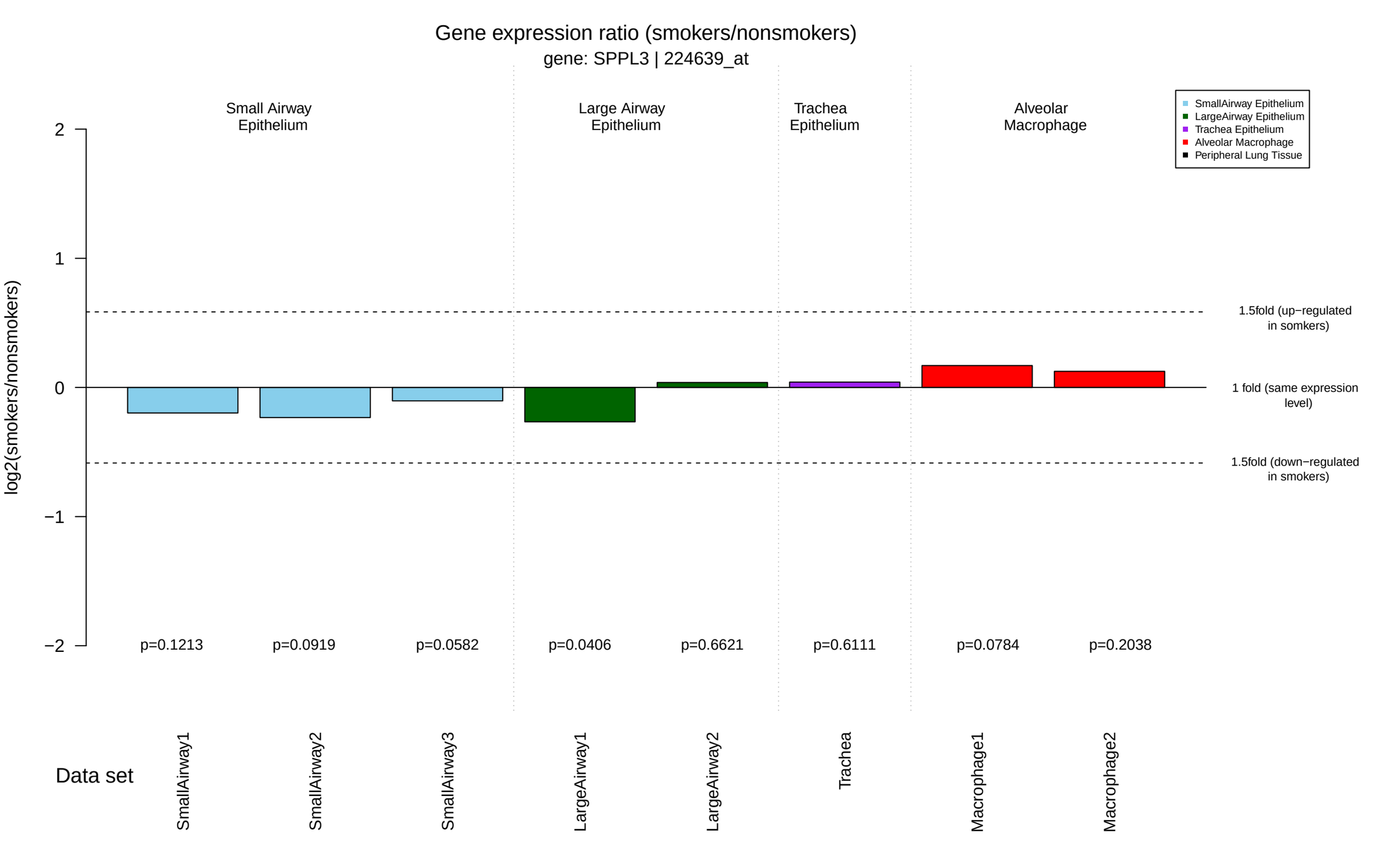
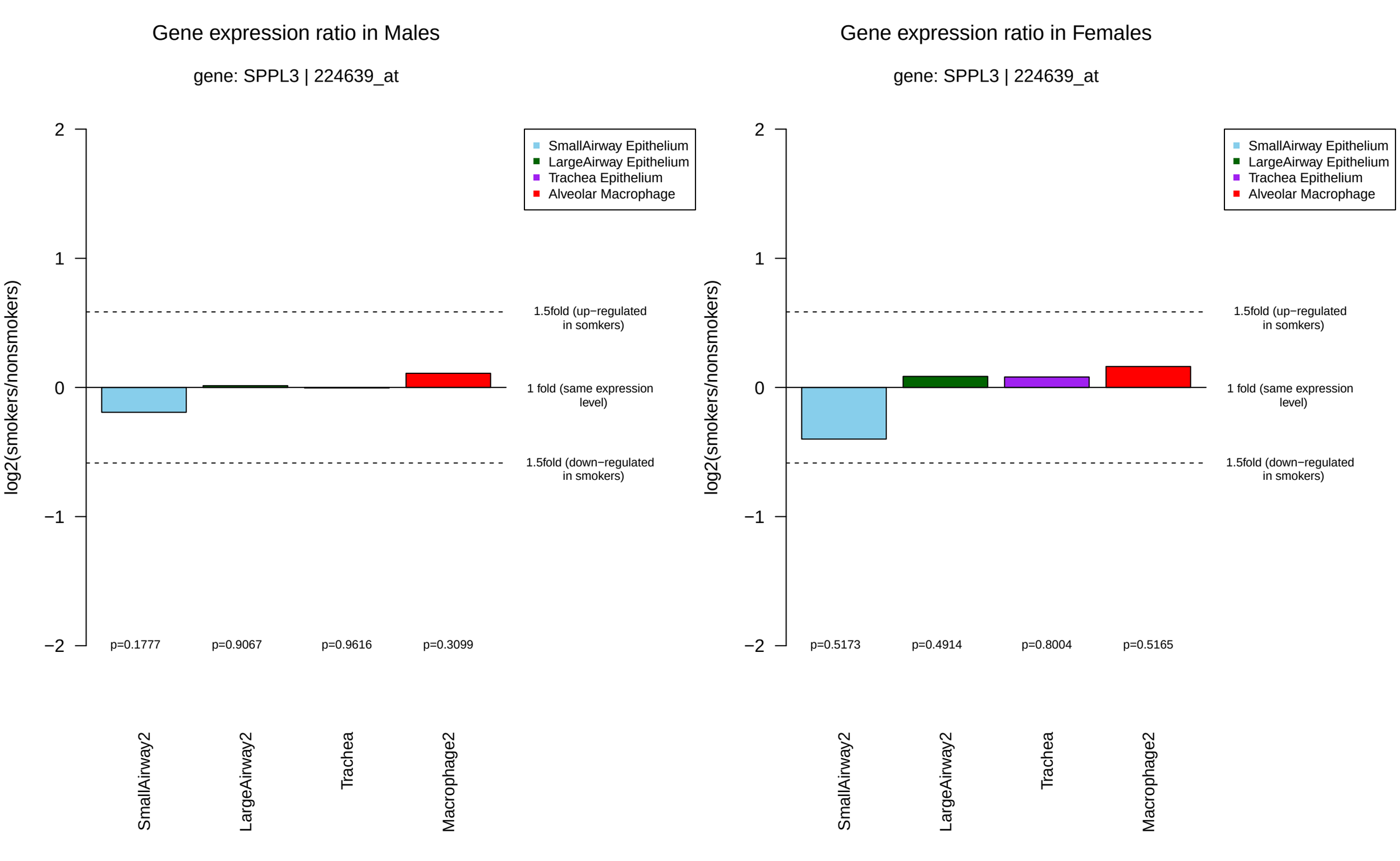
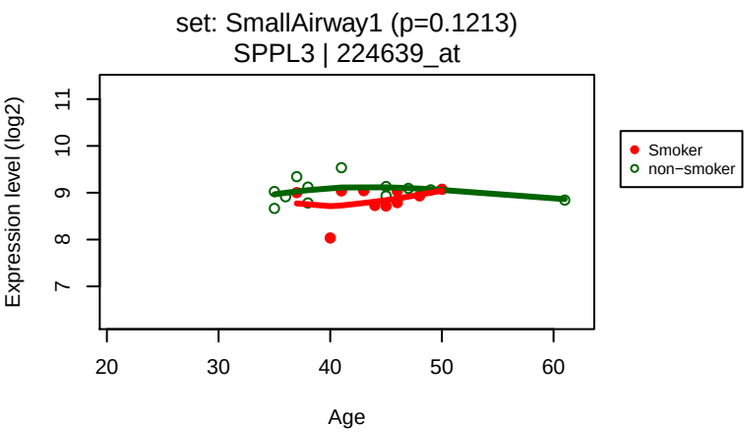
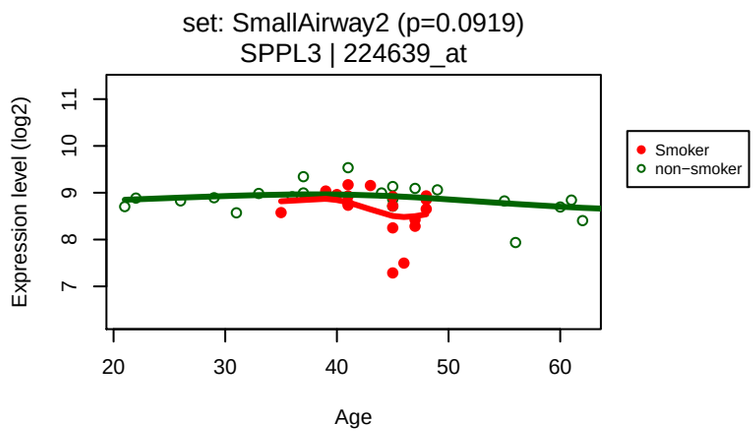
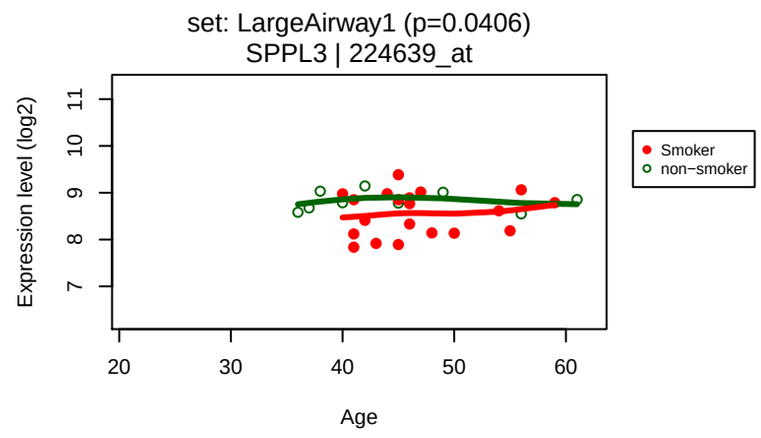

Gene expression of smokers and nonsmokers
gene: SPPL3 | 224639_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1213
p=0.0919
p=0.0582
p=0.0406
p=0.6621
p=0.6111
p=0.0784
p=0.2038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

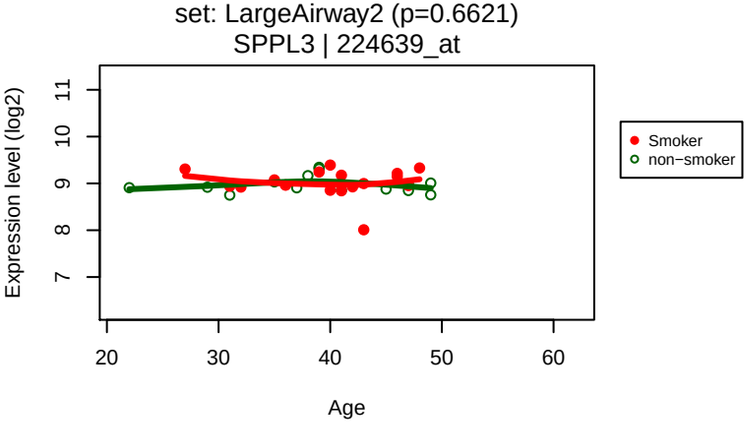

Gene expression ratio (smokers/nonsmokers)
gene: SPPL3 | 224639_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1213
p=0.0919
p=0.0582
p=0.0406
p=0.6621
p=0.6111
p=0.0784
p=0.2038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

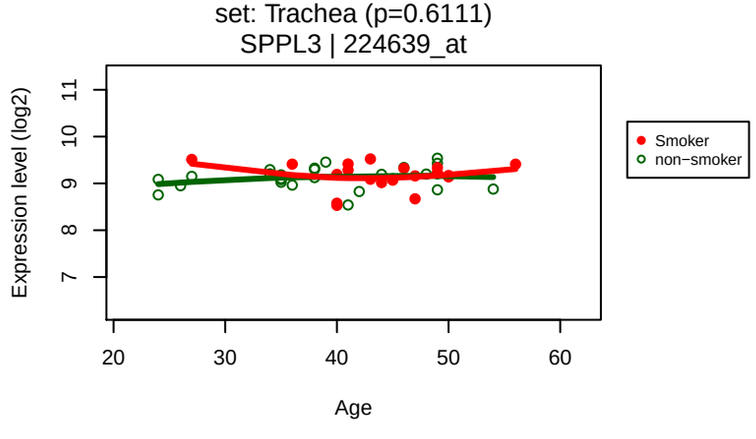

Gene expression ratio in Males
gene: SPPL3 | 224639_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1777
p=0.9067
p=0.9616
p=0.3099
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPPL3 | 224639_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5173
p=0.4914
p=0.8004
p=0.5165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

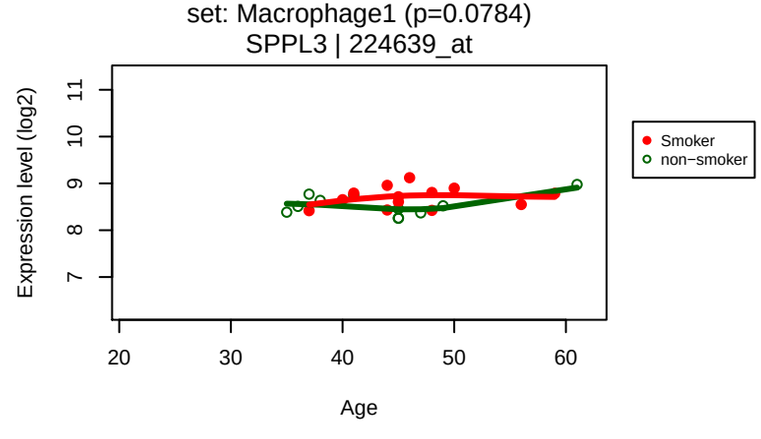

set: SmallAirway1 (p=0.1213)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker

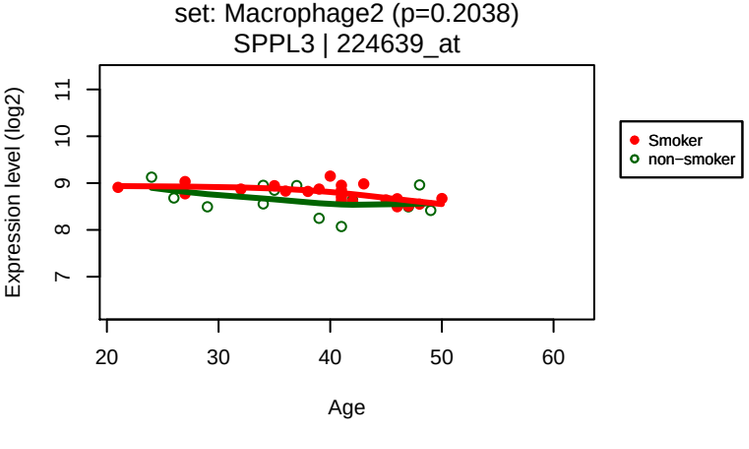

set: SmallAirway2 (p=0.0919)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker

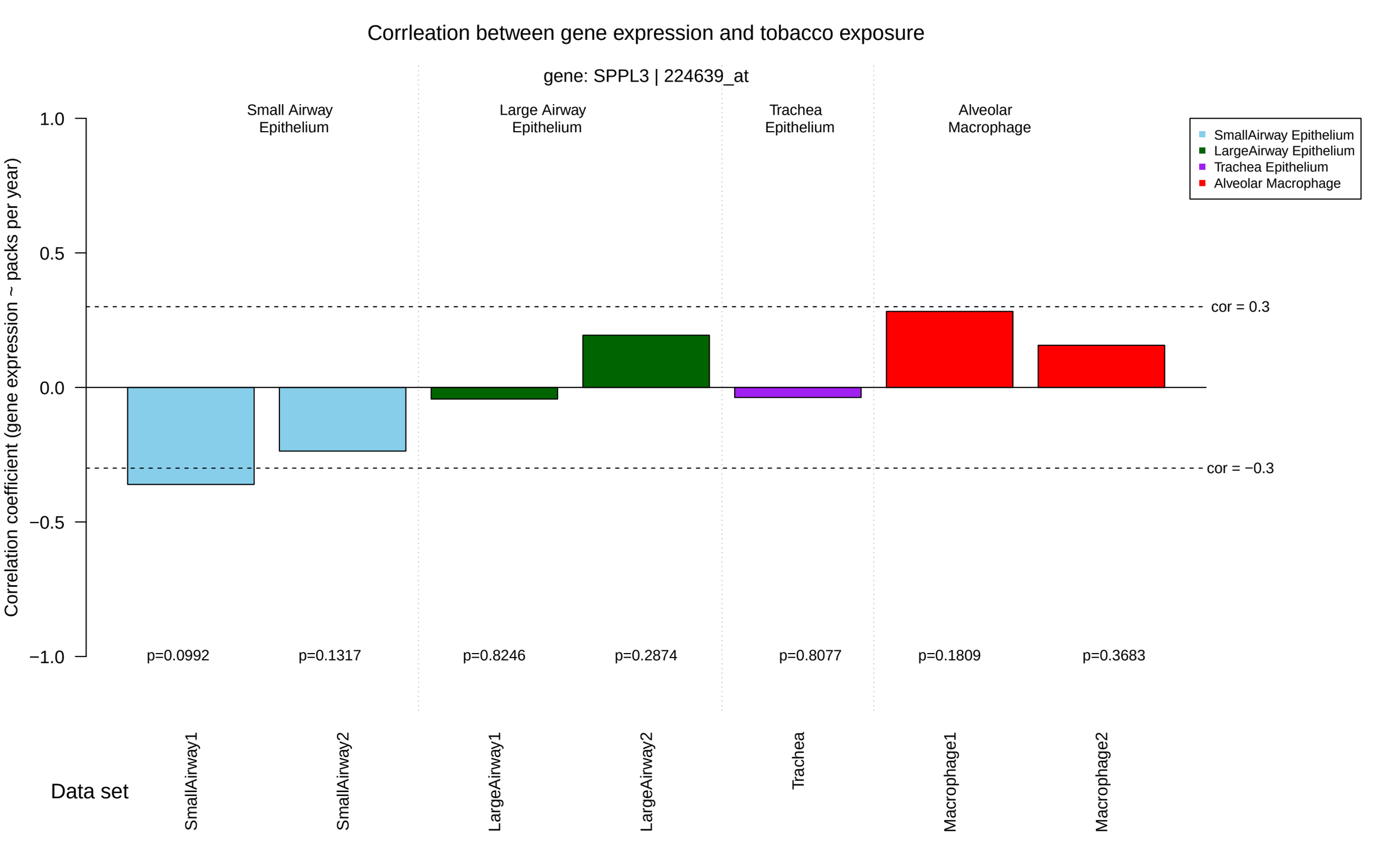

set: LargeAirway1 (p=0.0406)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6621)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6111)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0784)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2038)
SPPL3 | 224639_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPPL3 | 224639_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0992
p=0.1317
p=0.8246
p=0.2874
p=0.8077
p=0.1809
p=0.3683
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2